

A Diode Model of KOSPI Market Dynamics: Refuting the RC Monotonic-Recovery Hypothesis with a 2-Year Backtest

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2026-06-26 · Research purpose only · Not investment advice · Data: Naver Finance (KOSPI daily price & investor-type net flows)

Abstract. We identify why the prior RC (resistor-capacitor) monotonic-recovery model fails structurally during the June 2026 KOSPI crash-recovery episode, and propose an **RLC + diode** model. The central hypothesis is a directional diode **D_sell = (foreign net-sell AND institutional net-sell)** that ignites market crashes. In a backtest over **604 trading days** (2024–2026), days with D_sell ON had a mean daily return of **-1.19%** versus **+0.68%** when OFF, a 1.87 pp gap with an overwhelming **t = 9.55**. The diode captured 83% of crash days ($\leq -5\%$), and its directional classification accuracy on large-move days ($\geq 3\%$) was 82.4%. We honestly report that D_sell is contemporaneous with the close, so it has **no next-day point-forecast power** (mean reversion dominates). The RC model cannot simultaneously explain the rallies of Jun 24–25 (rising amid foreign selling) and the Jun 26 crash (joint selling); the diode model classifies both with statistical significance.

1. Introduction — Structural Failure of RC Monotonic Recovery

Prior RC models (v3, v7) described KOSPI recovery as a driven RC relaxation:

$$C_{\text{eff}} \cdot dV/dt = \sigma_{\text{total}} \cdot (V_{\text{target}} - V) - I_{\text{forced}}$$

solution (no forced sell): $V(t) = V_{\text{target}} - (V_{\text{target}} - V_0) \cdot e^{-(t/\tau)}$

This form forces V to converge **monotonically toward V_{target}** . Yet the **one-day reversal** from 2026-06-25 (+5.42%, 8,930) to 06-26 (-5.81%, 8,411) is irreproducible by a monotonic solution.

RC omits three physical quantities:

- **(i) Inertia (L):** program/leverage selling carries momentum; RC has no second-order term.
- **(ii) Directional asymmetry (diode):** joint foreign+institutional selling differs qualitatively from foreign-only selling.
- **(iii) Foreign-gate fallacy:** RC v7 treats the foreign sign as master switch, but Jun 24–25 rallied during foreign selling.

2. The Diode Model

We add a directional diode in series with an RLC circuit:

$$L \cdot d^2V/dt^2 + R \cdot dV/dt + (V - V_{\text{target}})/C = -D_{\text{sell}}(t) \cdot K \cdot |F+I|/C_{\text{flow}}$$

$$D_{\text{sell}}(t) = 1 \quad \text{iff} \quad F(t) < 0 \quad \text{AND} \quad I(t) < 0 \\ = 0 \quad \text{otherwise}$$

F, I are foreign/institutional daily net flows. The **diode conducts only during joint (panic) selling**, igniting the shock term. For damping ratio $\zeta < 1$ the system is underdamped, naturally producing the **overshoot/oscillation** that RC's monotonic convergence cannot.

Empirical calibration: $C_{\text{flow}} = 0.017$ tril/pt, $K_{\text{cascade}} = 1.78$ (D+0 cascade fit), $\tau = 2.32$ days.

3. Data and Method

KOSPI daily closes (2024-01-02 to 2026-06-26, 604 days) and investor-type daily net flows were collected from Naver Finance. D_{sell} is set from the foreign/institutional net-flow signs. The backtest comprises four hypotheses: H1 (contemporaneous classification), H2 (crash capture), H3 (next-day forecast, ex-ante), H4 (trading, prior-day signal, no look-ahead).

4. Backtest Results

4.1 H1 — Contemporaneous Classification (core result)

State	Days	Mean %	Median %	Down-day %
D_{sell} ON	151	-1.19	-0.84	80.1%
D_{sell} OFF	453	+0.68	+0.49	29.4%

Gap 1.87 pp, **t = 9.55**. The foreign sign alone cannot explain the Jun 24–25 rallies; the **institutional sign (I) is the true switch**.

4.2 H2 — Crash Capture

Threshold	Days	D_{sell} ON	Capture
$\leq -1\%$	94	66	70.2%
$\leq -2\%$	40	32	80.0%
$\leq -3\%$	26	20	76.9%
$\leq -5\%$	12	10	83.3%
$\geq +2\%$ (control)	61	5	8.2%

Larger crashes yield higher capture; the diode rarely fires on up-days, confirming **directional asymmetry**.

4.3 H3 — Next-Day Forecast (honest limitation)

Using prior-day D_sell to forecast next-day returns yields a -0.14 pp gap — **forecast power vanishes and reverts**. D_sell is contemporaneous with the close, so it is too late once the close is fixed. Real value arises only from **intraday real-time flow tracking**.

4.4 H4 — Trading Backtest

Strategy	Total %	CAGR %	Sharpe	MDD %
Buy & Hold	+215.0	+61.4	1.68	-20.7
RC (buy day after drop)	+96.6	+32.6	1.32	-18.4
Diode (avoid on ON)	+105.9	+35.2	1.33	-19.4
Diode (short on ON)	+23.2	+9.1	0.43	-34.8

2024–26 was a 4× bull market; avoidance cannot beat Buy & Hold. The diode's value is in **diagnosis/alerting**, not trading alpha.

5. Scorecard vs RC/Codex (Before / After Model Change)

Criterion	Before (RC v3 / Codex v7)	After (Diode v5)	Win
Jun 26 direction	Predicted 9,030 → actual 8,411 (✗)	D_sell ON crash captured (✓)	After
Explaining Jun 24–25 rally	Rose amid foreign selling = unexplained	Institution buy (I>0) → OFF (✓)	After
Contemporaneous classification	Qualitative σ_{total} narrative	t = 9.55 (quantified)	After
Crash capture ($\leq -5\%$)	No such concept	83% captured	After
US-signal collection	Multi (SOXX·EWY·MU), systematic	Single V_target proxy	Before
Next-day ex-ante point forecast	Probabilities published (attempted)	Limitation acknowledged	Draw
Intellectual honesty	Probabilities disclosed	Limits disclosed	Draw

Overall: the model structure (After/diode) clearly surpasses RC in contemporaneous classification and crash capture. Codex leads on systematic US-signal collection, a target for v6 integration. **Codex continues to compete actively with RC v7; this model is the structural response to it.**

6. Limitations and Conclusion

Limitations: (1) no next-day point forecast — intraday data required; (2) single market, 2.5-year sample; (3) K_cascade/C_flow calibrated in-sample; (4) trading costs excluded.

Conclusion: the RC monotonic-recovery assumption is physically refuted. The diode model classifies market direction at $t = 9.55$ over 604 days and demonstrates that the institutional sign is a superior switch to the foreign gate. The next step (v6) is to capture the **D_sell ignition timing** from intraday 30-minute cumulative foreign/institutional net flows, and to integrate Codex's multi-US-signal scheme into V_target.

KOSPI Diode Model v5 · Claude × Cheom-mul · code/data: github.com/waterfirst/kospi-mw-rc-research
Live simulator: deployed on GitHub Pages · Research purpose only, not investment advice · Costs/taxes/slippage excluded.